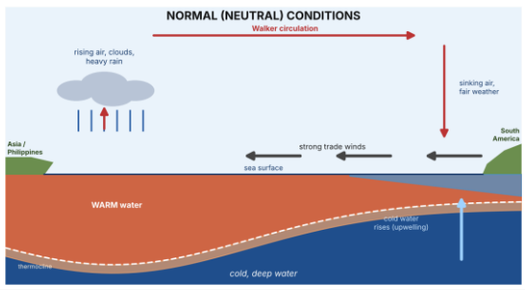
DAILY LESSON PLAN	School	**Taguig National High School**	Grade Level	**10**
	Teacher	**Ian B. Corocoto**	Learning Area	**SCIENCE (Earth Science)**
	Teaching Date and Time	**Day 1** **45 minutes**	Quarter	**Second Term (Week 9)**

I. OBJECTIVES		
A. Content Standards	Climate change and its impacts on the environment and people pose serious challenges which require solutions and action at local and global levels.	
B. Performance Standards	By the end of the term, learners gather information from secondary sources to describe rapid changes that are occurring in local and global climate patterns and propose solutions to address these changes at the local and global levels by drawing on awareness, responsible personal behavior to conserve materials and energy, and through the better societal management of the natural resources of the country.	
C. Most Essential Learning Competencies/ Objectives	At the end of the teaching-learning process, 100% of the students with at least 80% proficiency are expected to: 1. describe the normal (neutral) conditions of the tropical Pacific that set the stage for ENSO, including the trade winds, the warm pool, upwelling and the Walker circulation.	
II. CONTENT	**Global Interactions: Global Climatic Phenomena – Introducing the El Niño Southern Oscillation (ENSO) and Normal (Neutral) Conditions**	
III. LEARNING RESOURCES		
A. References	DepEd Grade 10 Science Budget of Work, Second Term, Week 9 – Global Interactions: "Describe how global climatic phenomena, such as the El Niño Southern Oscillation, may impact weather systems." PAGASA (https://www.pagasa.dost.gov.ph) – ENSO and climate information NOAA Climate.gov – "What is ENSO?" and "El Niño & La Niña" (https://www.climate.gov/enso) Materials: laptop and projector, diagram "Normal (Neutral) Conditions", W/C response cards, shallow tray, water, straw, ruler, food coloring, paper bits, activity sheet	
IV. PROCEDURES	**Teacher's Activity**	**Student's Activity**
	Classroom Conditioning (Let the teacher count 1, 2, 3)	(As the teacher counts, the students will: 1- arrange the chairs, 2- pick up the pieces of paper, and 3- be ready to pray)
	Prayer Let us pray.	(The students will pray.)
	Greetings Good morning, class! Please be seated.	Good morning, Sir! It's nice to see you. Thank you.
	Checking of Attendance Who is absent for today?	The class secretary will report the attendance for today.
A. Reviewing previous lesson or presenting the new lesson (ELICIT) *5 minutes*	Before we begin, let us play **"WEATHER or CLIMATE?"** **Mechanics:** I will read a statement. Raise the **W** card if it describes WEATHER and the **C** card if it describes CLIMATE. You have 5 seconds for each. Are you ready? Let's begin!	Yes, Sir!
	1. "It is raining in Taguig right now." 2. "The Philippines is hot and humid most of the year." 3. "Signal No. 2 was raised over Samar this morning." 4. "Baguio is cooler than Manila throughout the year."	1. W 2. C 3. W 4. C 5. W (The students raise their cards.)

	5. "Yesterday's heat index reached 41 °C."	
	Very good! So, based on the game, what is the difference between weather and climate?	Weather is the condition of the atmosphere in a place at a certain time, like rain today or heat this afternoon. Climate is the average weather of a place over a long period of time, usually about 30 years.
	Excellent! Weather changes from hour to hour, while climate is the long-term pattern. Now, can weather in the Philippines be influenced by something that happens far away, like thousands of kilometers across the ocean? That is what we will explore for the next four days: **how global climatic phenomena affect our weather.**	Maybe, Sir. (Answers may vary)
B. Establishing a purpose for the lesson (ENGAGE) *4 minutes*	Let us look at these headlines that I prepared: ☐ "Dams drop to low levels; water rationing in some areas" ☐ "Heat index reaches danger level; classes suspended or moved online" ☐ "Farmers lose crops as rains fail to arrive" ☐ "Floods and landslides after weeks of heavy rain" Have you or your family experienced any of these?	Yes, Sir. Last year it was very hot and some classes were suspended. Our neighborhood had little water for days. (Answers may vary)
	Thank you for sharing. These events are different, but scientists often link them to a pattern in a place **very far from here**, the Pacific Ocean. What do you think happens in the ocean that can make us too dry in one year and too wet in another?	Maybe the sea gets warmer or colder, Sir. (Answers may vary)
	Great guesses! Today, you will find out how the ocean and the air work together. Our lesson objective is to: **describe the normal (neutral) conditions of the tropical Pacific**, including the trade winds, the warm pool, upwelling and the Walker circulation. This is the starting point for understanding ENSO.	(The students read the objective silently.)
C. Presenting examples/ instances of the new lesson (EXPLORE) *10 minutes*	Before we study the real Pacific Ocean, let us make a small model of it. **ACTIVITY: "WIND PUSHES WATER"** (Group of 5; 10 minutes; activity sheet provided) **Objective:** To observe how wind affects the movement and level of water. **Materials:** long shallow tray or plastic container, water, drinking straw, ruler, food coloring, small bits of paper, marker **SAFETY PRECAUTIONS:** Wipe spills right away to prevent slipping. Blow gently through the straw; do not suck water into it. Use your own straw only.	(The students gather their materials and listen to the instructions.)
	Procedure: 1. Pour water about 2 cm deep into the tray. Add one drop of food coloring. 2. Sprinkle bits of paper on the water at the center. 3. With the marker, mark the water level at BOTH ends of the tray. 4. Blow **gently** across the water from one end for 30 seconds. Record what the paper bits do and what happens to the water level at the far end.	No, Sir. (The students perform the activity and record their observations in the data table.)

	5. Rest for one minute, then blow **harder** for 30 seconds. Record again. 6. Answer the guide questions. Do you have any clarifications? You may now begin.	
	(The teacher moves around, asks guiding questions, and checks that every member is participating.) • What do you notice about the paper bits? • Where does the water seem deeper?	The paper bits move in the direction of the wind. The water looks higher at the far end.
D. Discussing new concepts and practicing new skills #1 (EXPLAIN) *10 minutes*	Time is up! Let us present the findings of each group. (Two groups report.) **Guide Questions:** 1. What happened to the paper bits when you blew across the water?	The paper bits moved in the same direction as the wind.
	2. What happened to the water level at the far end? Was there a difference when you blew harder?	The water piled up at the far end. When we blew harder, the pile was higher and moved farther.
	3. If the tray were the Pacific Ocean, what would the wind represent and what would the piled-up water represent?	The wind is like the trade winds and the piled-up water is like warm water gathering in one side of the ocean.
	Excellent! You have just discovered the key idea of today's lesson: **wind can push surface water and make it pile up.** Now let us look at the real Pacific Ocean in **normal (neutral) conditions.** 	(The students observe the diagram.)
	Look at the diagram. Which direction do the **trade winds** blow along the tropical Pacific?	From east to west, from South America toward Asia, Sir.
	Correct! The trade winds push the warm surface water westward. Where do you think the warm water gathers?	In the western Pacific, near Asia and the Philippines.
	Yes! We call this the **WARM POOL**. It is the part of the ocean with the warmest surface water in the world. Warm water evaporates easily, so what happens to the air above it?	The warm, moist air rises and forms clouds, so there is more rain.
	Very good! Rising air over the warm pool brings clouds and heavy rain to the Philippines and Indonesia. The air then travels east high in the sky, sinks over the eastern Pacific (fair weather there), and returns west as trade winds. This loop is the **WALKER CIRCULATION**. What about the eastern Pacific near South America? Because the trade winds push surface water away, what rises to take its place?	Cold water from below rises up, Sir.
	Exactly! That is called **UPWELLING.** The cold water is rich in nutrients, so Peruvian fishermen catch many fish, especially anchovies. Now, we call the **boundary between the warm surface layer and cold deep water the THERMOCLINE.** In normal conditions it is deep in the west and shallow in the east.	(The students copy the key terms in their notebooks.)
E. Developing mastery	Now, the pattern in the Pacific does not stay the same all the time. Sometimes the trade winds	(The students listen and take notes.) Wow, Sir! It is named after Christmas!

(leads to Formative Assessment) (EXPLAIN) *3 minutes*	weaken and sometimes they strengthen. This irregular swing is called the **El Niño Southern Oscillation, or ENSO.** **ENSO**, as defined by scientists, is a natural cycle in which the sea-surface temperatures of the central and eastern tropical Pacific warm (**El Niño**), cool (**La Niña**), or stay near normal (**neutral**), together with changes in winds and rainfall. It repeats irregularly about every 2 to 7 years. Trivia: "El Niño" is Spanish for "the Christ child." Peruvian fishermen noticed that the sea became warm around Christmas time.	
	Is ENSO caused by humans?	No, Sir. It is a natural cycle.
	Correct! It is natural. Scientists are still studying how global warming may be changing its effects, but ENSO existed long before humans started burning fossil fuels.	(The students nod.)
F. Finding practical application of concepts and skills in daily living (ELABORATE) *4 minutes*	Let us apply what you learned. **"WHERE WILL IT RAIN?"** Looking again at the neutral-conditions diagram, which side of the Pacific usually gets the heaviest rain, and why does this matter to the Philippines?	The western Pacific, because the warm pool is there. The Philippines is on the western side, so we usually receive plenty of rain.
	Why might farmers, fishermen, and water companies in the Philippines care about what happens in the middle of the Pacific?	If the pattern changes, the rain may become too little or too much, which can affect crops, fishing, and the water supply.
	Very good! That is why PAGASA monitors the Pacific Ocean every day. Tomorrow, we will see what happens when the normal pattern changes.	(The students listen.)
G. Making generalizations and abstractions about the lesson (ELABORATE) *3 minutes*	Let us summarize using **"Complete the Thought."** Complete the statements: 1. In normal conditions, the trade winds blow from _____ to _____. 2. The _____ is the area of warm water near Asia, and _____ is the rise of cold water near South America. 3. Rising air over the warm pool brings _____ to the Philippines. 4. The loop of air over the tropical Pacific is called the _____.	1. East to west. 2. Warm pool; upwelling. 3. Clouds and heavy rain. 4. Walker circulation.
H. Evaluating learning (EVALUATE) *5 minutes*	**MULTIPLE CHOICE:** Read each question carefully. Write the letter of the correct answer on a ¼ sheet of paper. 1. The winds that blow from east to west along the tropical Pacific are called: a) monsoon winds b) trade winds c) jet streams d) land breezes 2. In normal conditions, where is the warm pool located? a) Near South America b) Near the North Pole c) In the western Pacific near Asia and the Philippines d) In the Atlantic Ocean 3. Why is the cold-water upwelling near South America important? a) It melts glaciers. b) It brings nutrient-rich water that supports fishing. c) It causes typhoons in the Philippines. d) It stops the trade winds.	1. B 2. C 3. B
	4. In normal conditions, why does the Philippines	4. A

	usually receive plenty of rain? a) Warm water near the Philippines makes moist air rise and form rain clouds. b) The Philippines is far from the ocean. c) Cold water rises near the Philippines. d) The trade winds blow all the clouds away. 5. Suppose the trade winds suddenly became very weak. What would most likely happen to the pile of warm water in the western Pacific? a) It would grow bigger. b) It would stay exactly the same. c) It would spread back east. d) It would disappear completely.	5. C (This prepares us for tomorrow's lesson on El Niño.)
I. Additional activities for application or remediation **(EXTEND)** *1 minute*	**Assignment:** 1. Ask a parent or an elder about a year when it was unusually dry or unusually rainy in your community. What happened to water, crops, or fishing? Write 3–4 sentences. 2. Read about **El Niño** and be ready to explain, in your own words, what could change in the Pacific if the trade winds weaken. 3. Bring colored pencils for tomorrow's activity. Thank you, class! Goodbye.	(The students write the assignment in their notebooks.) Goodbye and thank you, Sir!
J. REFLECTION		
A. No. of learners who earned at least 80% in the evaluation		
B. No. of learners who require additional activities for remediation		
C. Did the remedial lessons work? No. of learners who have caught up with the lesson		
D. No. of learners who continue to require remediation		
E. Which of my teaching strategies worked well? Why did these work?		
F. What difficulties did I encounter which my principal or supervisor can help me solve?		
G. What innovation or localized materials did I use/discover which I wish to share with other teachers?		

Prepared by:
IAN B. COROCOTO
Student Teacher

Checked:

[COOPERATING TEACHER]
Critic Teacher
Master Teacher 1

Noted:

[HEAD TEACHER]
Head Teacher III

Group: _____ Score: _____

"WIND PUSHES WATER"

Objective: To observe how wind affects the movement and the level of water.

Materials: long shallow tray, water, drinking straw, ruler, food coloring, bits of paper, marker

PRECAUTIONS: Wipe spills right away. Blow gently. Do not suck water into the straw. Use your own straw.

Procedure:

1. Pour water about 2 cm deep into the tray and add one drop of food coloring.
2. Sprinkle bits of paper at the center.
3. Mark the water level at both ends of the tray with a marker.
4. Blow gently across the water for 30 seconds from one end. Record.
5. Rest for one minute, then blow harder for 30 seconds. Record.

Trial	What the paper bits did	What happened to the water level at the far end?
Gentle wind		
Strong wind		

Guide Questions:

1. What happened to the paper bits when you blew across the water?
2. What happened to the water level at the far end? Did it change when you blew harder?
3. If the tray were the Pacific Ocean, what would the wind and the piled-up water represent?

Conclusion: _____

_____

Rubric for the Group Activity (20 points)

Criteria	4	3	2	1
Cooperation (everyone participates)	All members worked together	Most members worked together	Some members worked	Only one member worked
Procedure (followed steps and safety)	Followed all steps and safety rules	Missed one step	Missed two steps	Did not follow the steps
Accuracy of observations	Complete, accurate, detailed	Mostly accurate	Some are inaccurate	Mostly missing or wrong
Explanation (link to the Pacific)	Clearly links the model to the ocean	Partly links	Weak link	No link
Neatness and time	Neat; finished on time	Mostly neat	Messy	Incomplete

Answer Key – Day 1 Quiz

1. B 2. C 3. B 4. A 5. C